

BAAZ JKBOSE



2026 CLASS-11TH

Excretory Products and
Their Elimination

LECTURE-2

Zoology

Dr Adil Altaf sir



Topics To Be Covered

1

Human Excretory System

Revision

→ Ammoniotelia eg Bony fishes

→ Land Snails = Uricotelic

→ Heart Muscle Damage = Heart Attack.
(Sudden)

(d/w)

Gout

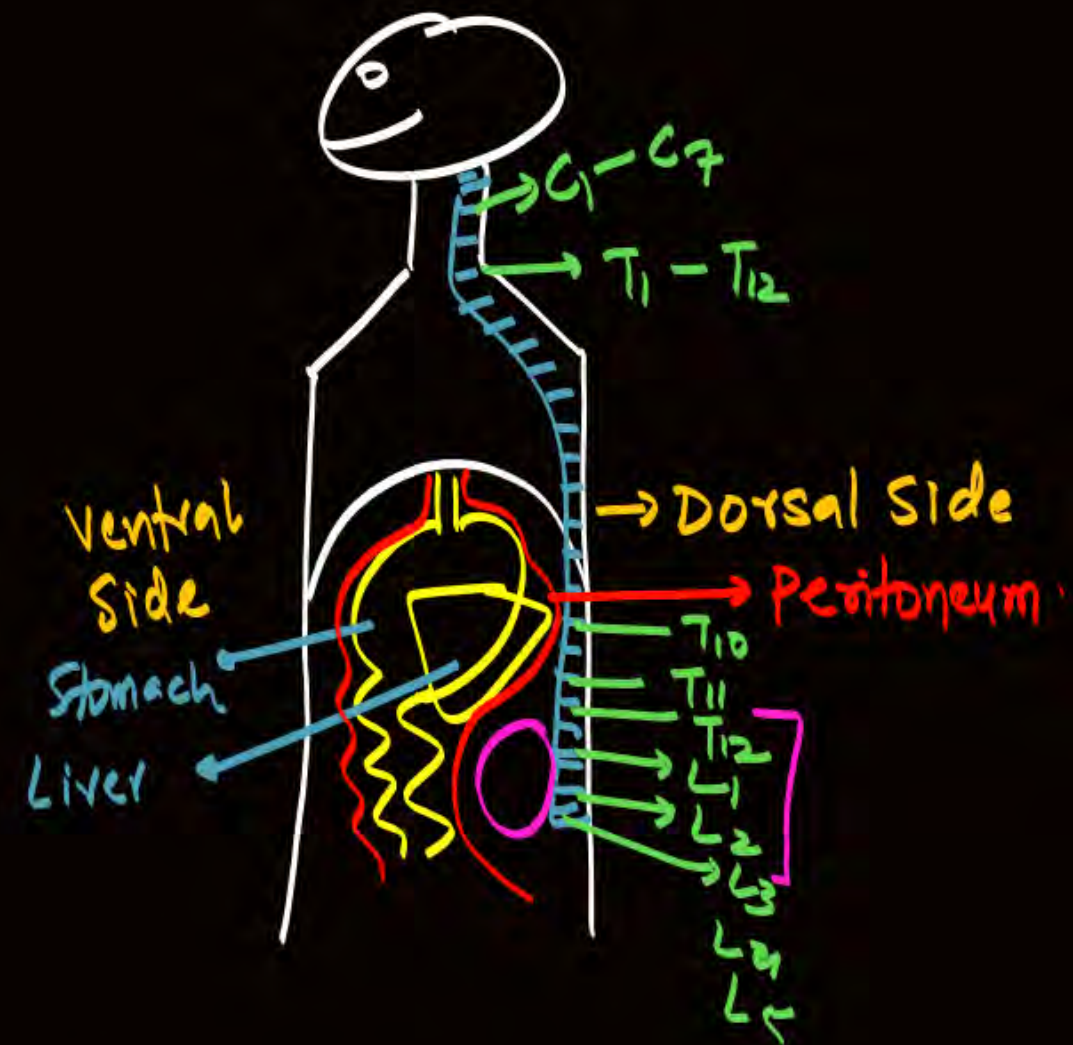
- Accumulation of Uric Acid
- In joints → Gouty Arthritis
- Disturbed purine metabolism.

Human Excretory System

↳ System involved in Excretion.

↳ It Includes :

- ✓ (i) Pairs of Kidney
- ✓ (ii) Pairs of Ureter
- ✓ (iii) Single Urinary Bladder
- ✓ (iv) Single urethra



(i) Kidney

Shape - These are Bean Shaped

Colour - Dark Brown in colour.

Dimension Length = 10-12 cm, Breadth = 5-7 cm, Thickness = 2-3 cm

Weight ⇒ 120 gm - 170 gm

→ location - located Dorsally in Abdominal cavity.

- Btw T₁₂ & L₃. (level) [Rt kidney is slightly lower than left]
(Thoracic 12) (3rd Lumbar).

→ Arrangement - It is Retroperitoneal
(Behind Peritoneum)

Internal Structure of kidney.

- ↳ Internally it has - outer cortex. (Deep pigmented)
- Inner Medulla (Light coloured)

↳ Covering of kidney is called - Renal Capsule

↳ Cortex is outer Region and Medulla is inner Region.
Medulla is present in the form of Medullary Pyramids.

↳ The columns present Btw Medullary Pyramids
k/as - Column of Bertini

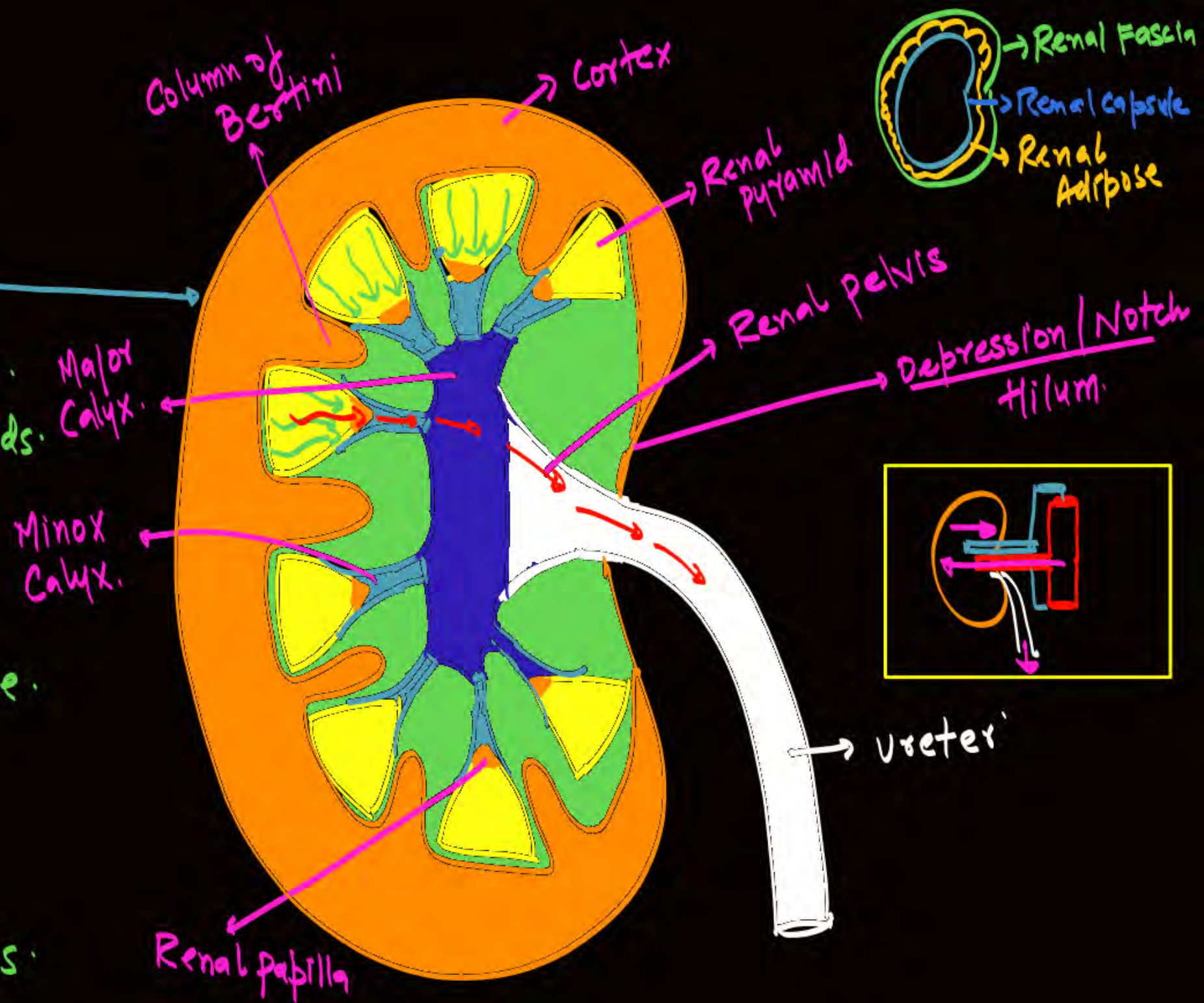
↳ Tips of Renal pyramids are called Renal papillae.

↳ These Renal Papillae open into Minor calyx.

↳ Minor Calyx form Major Calyx.

↳ Major Calyces open into Funnel Shaped k/as.
Renal pelvis.

↳ Renal pelvis continues with ureter.



II) Ureter (paired)

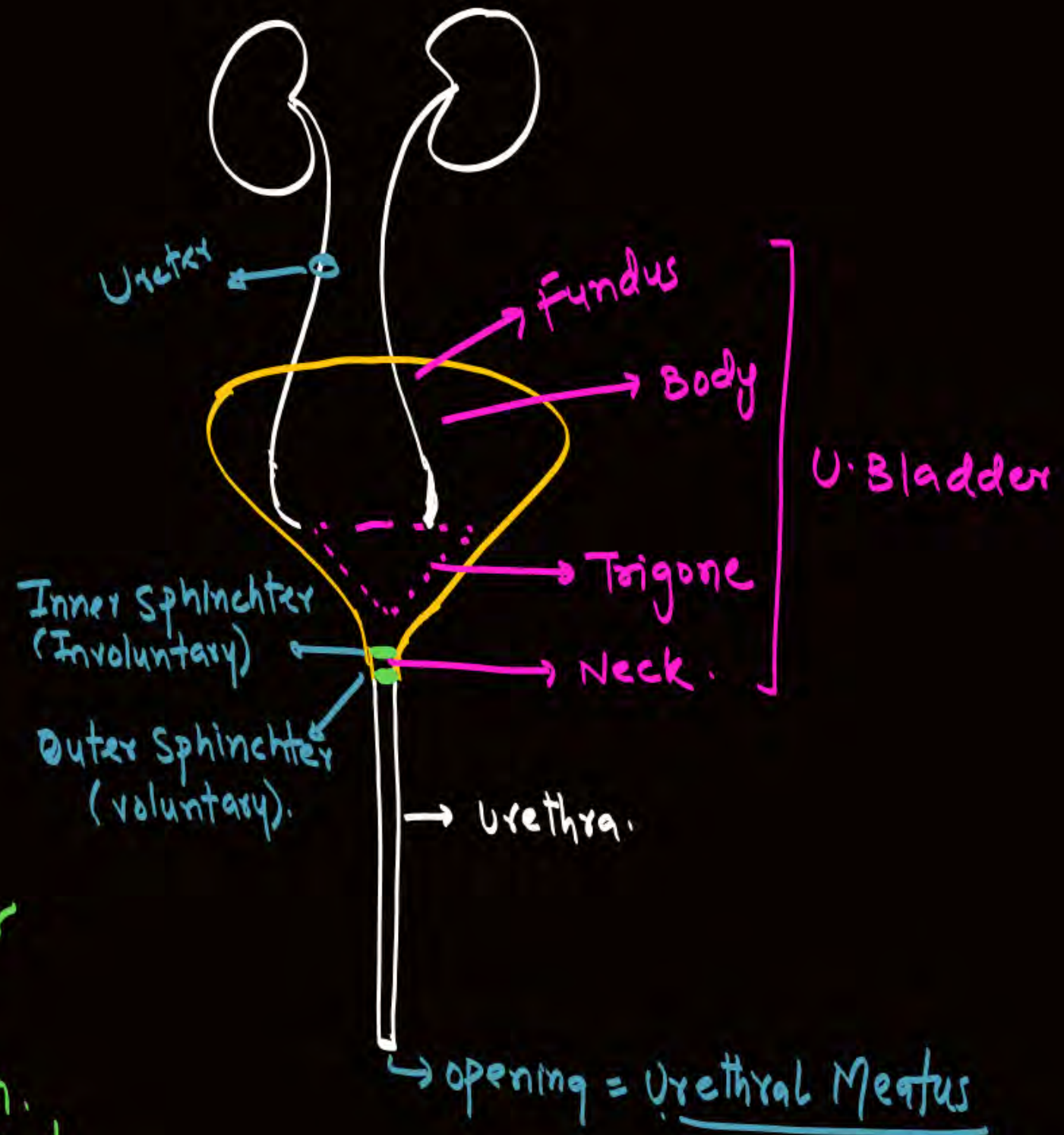
- These are Muscular tubes
- Connect pelvis of kidney to Urinary Bladder.
- Carry urine from kidney to U. Bladder By doing peristalsis.

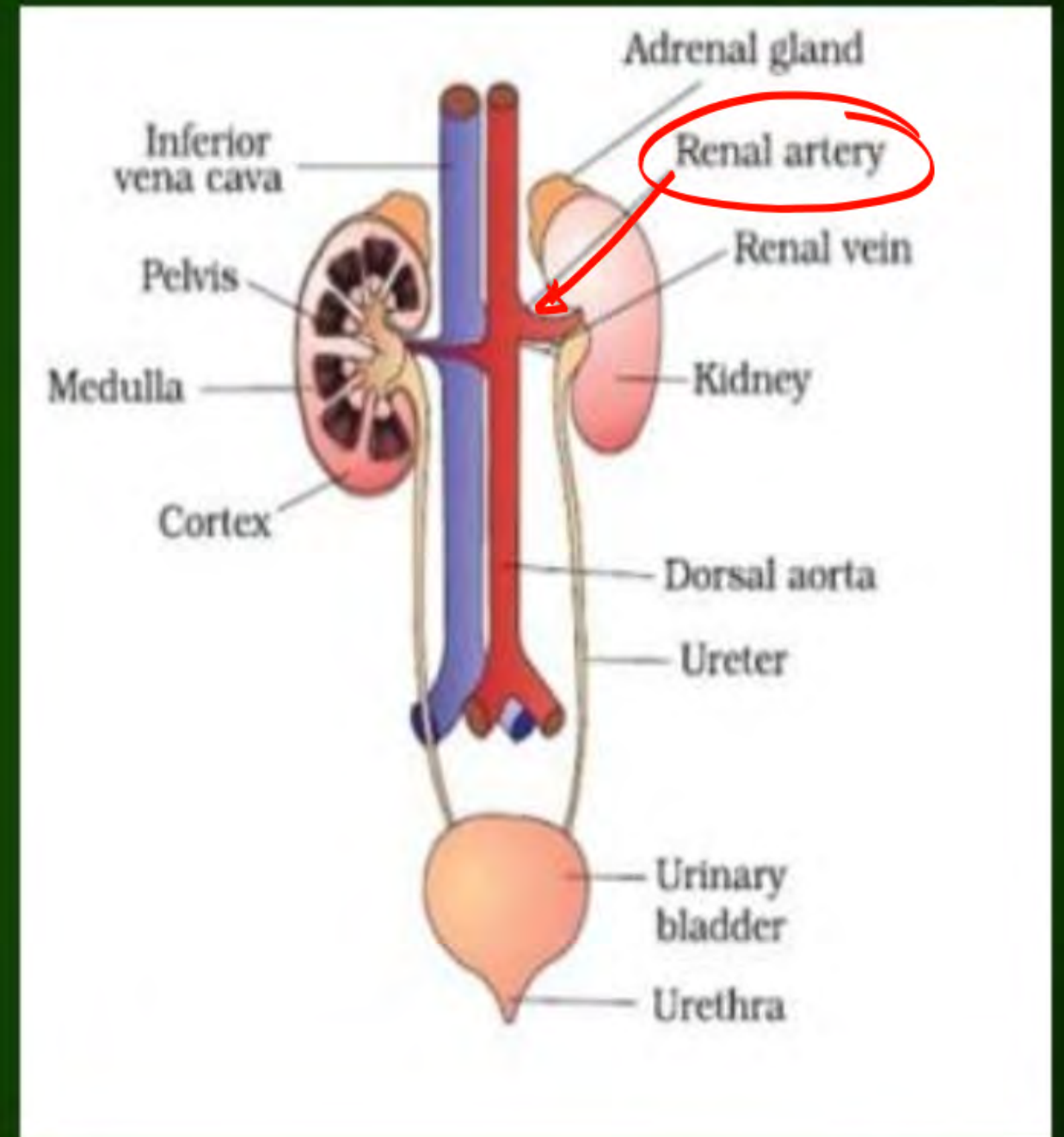
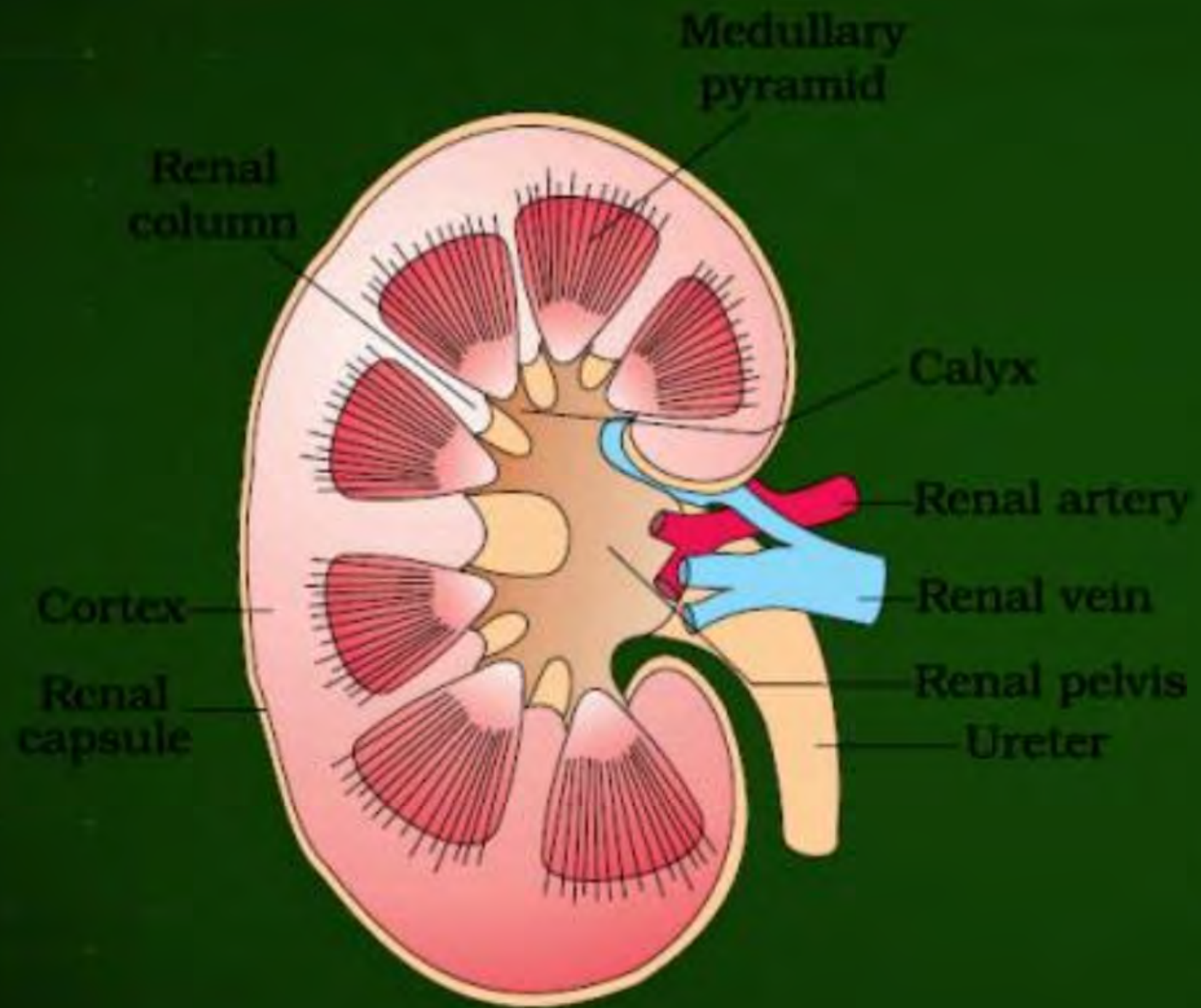
III) Urinary Bladder

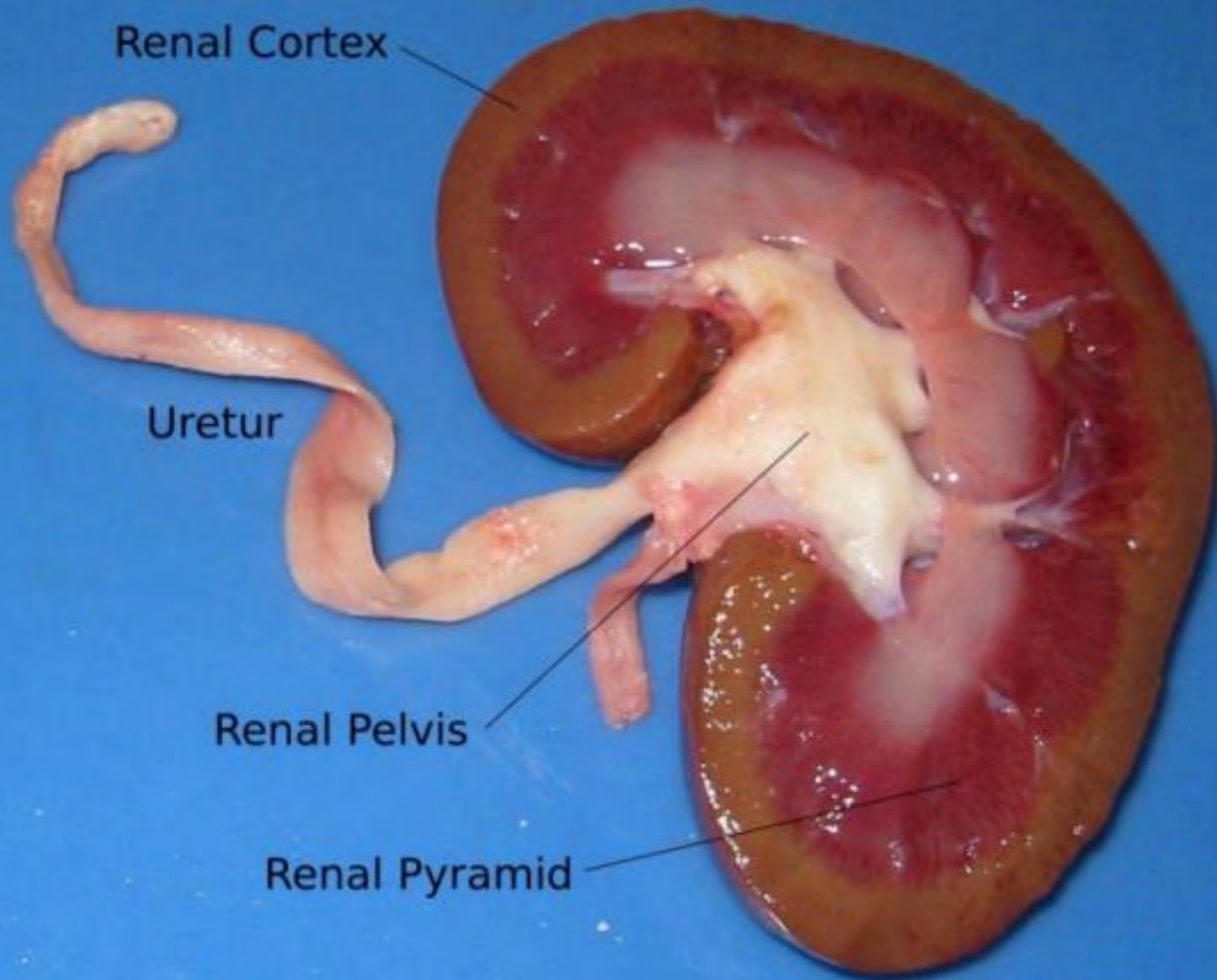
- Single Pouch made of Smooth Muscle (Involuntary).
- Used to hold Urine.
- Parts - Fundus, Body, Trigone, Neck.

IV) Urethra

- Muscular tube carries Urine out of U. Bladder
- Females (2-4cm) and carries only Urine.
- Males (25cm) and carries Urine as well Semen.
Hence klas Urinogenital Duct







Nephron

↳ It is the Structural and Functional Unit of kidney.

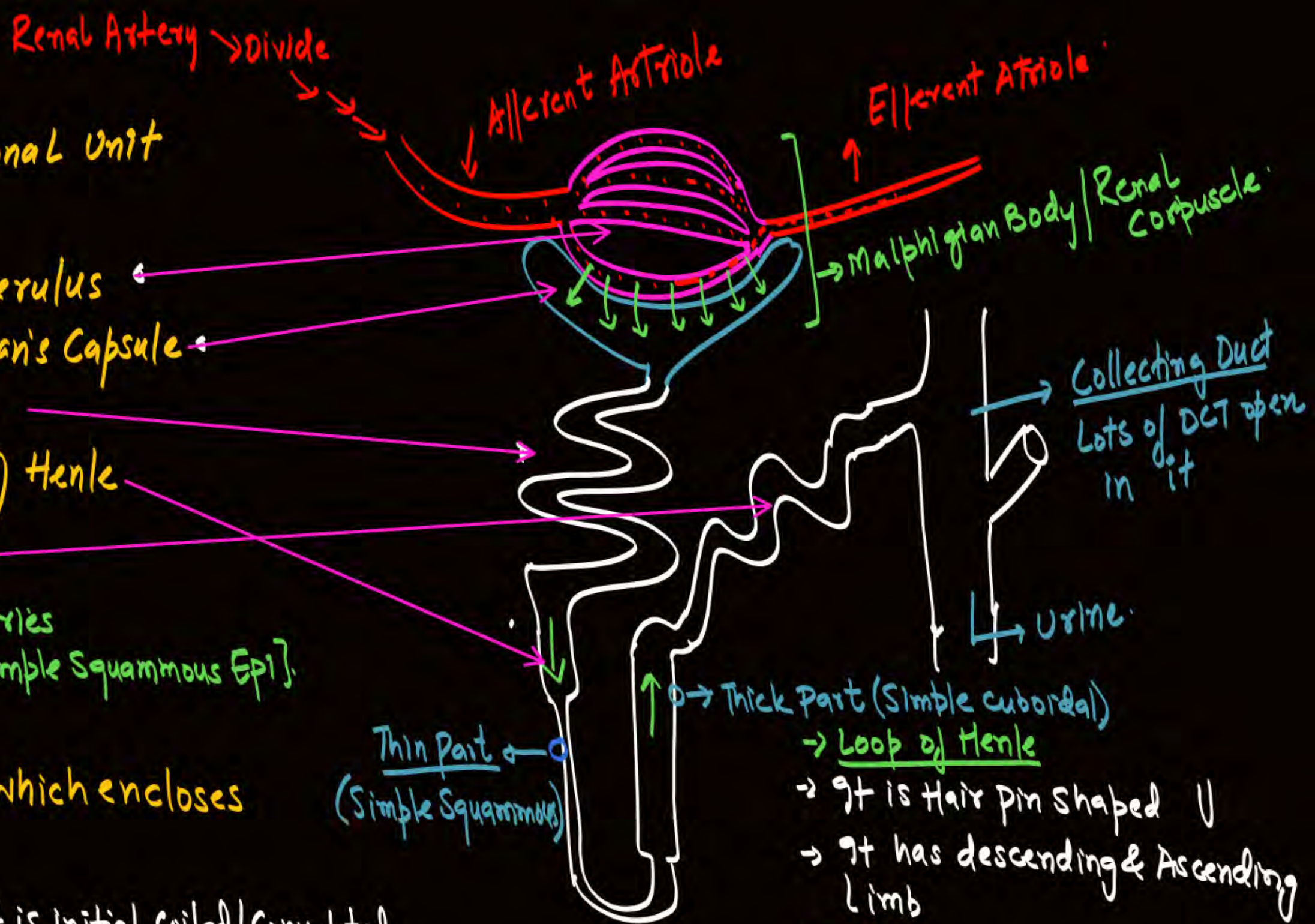
↳ Structurally Nephron has → Glomerulus
→ Bowman's Capsule
→ PCT.
→ Loop of Henle
→ DCT.

→ Glomerulus - Network/Tuft of Capillaries
- Made of Endothelium (Simple Squamous Epi).

→ Bowman's Capsule → It is that part which encloses Glomerulus

→ PCT. - Proximal convoluted tubule - It is initial coiled/convoluted part
- Made of Simple cuboidal Epithelium
- It Bears Microvilli - Increases Surface Area.

Microvilli



→ Thick Part (Simple cuboidal)

→ Loop of Henle

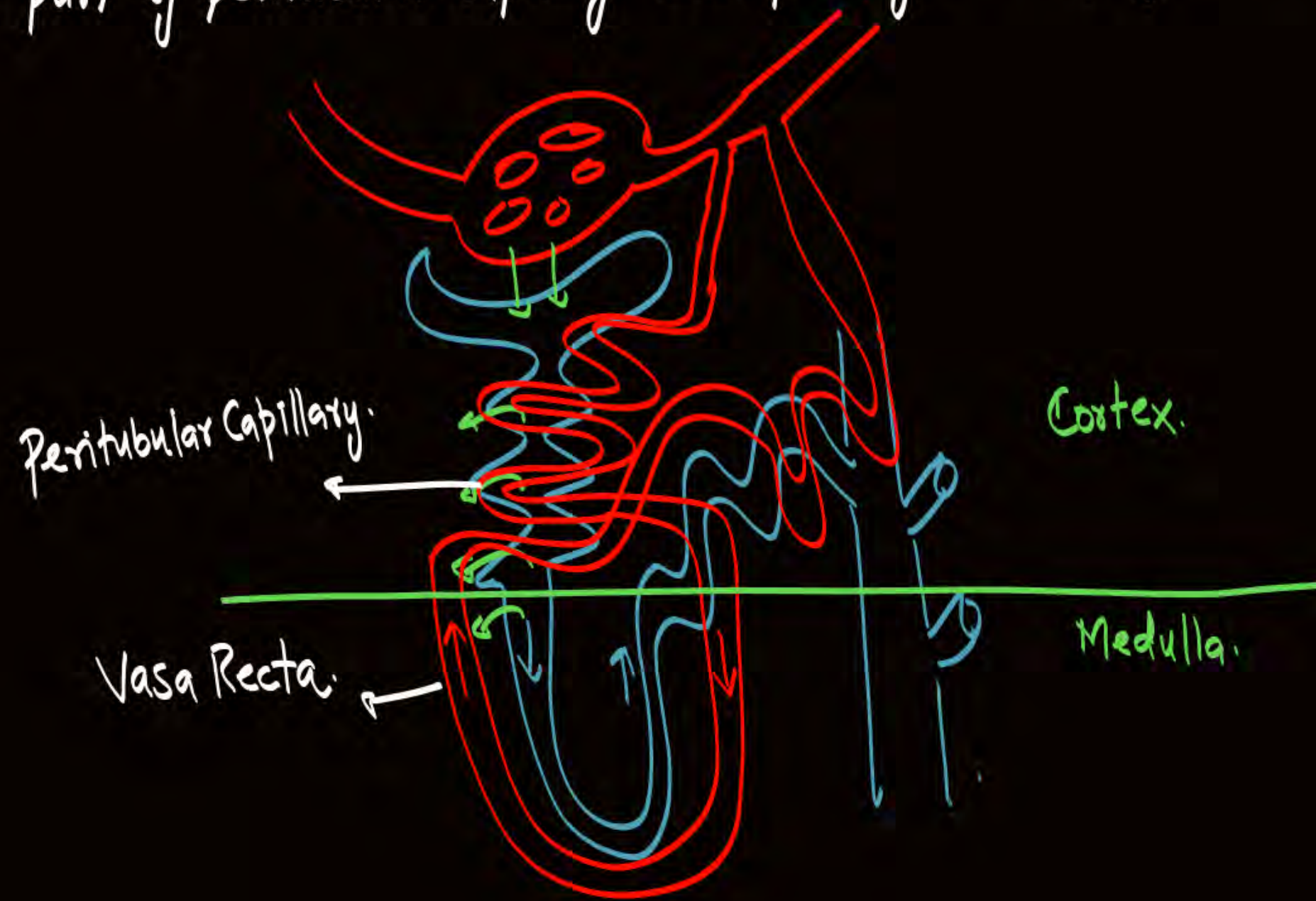
→ It is Hair pin shaped U

→ It has descending & Ascending Limb

→ DCT - Distal convoluted tubule.

→ Made of Simple cuboidal Epi.

- The contents which are Reabsorbed by kidney tubules are given into peritubular capillaries
- The part of peritubular capillary corresponding with loop of Henle is known as — Vasa Recta.



HOMWORK

Types of Nephron $\left\{ \begin{array}{l} \rightarrow \text{Cortical} \\ \rightarrow \text{Juxtamedullary} \end{array} \right.$

Insects
malpighian
tubules

Simple (Single layer)

Squamous



cuboidal



Thank
You